



Society of Petroleum Engineers

SPE-227664-MS

Advancing Impact Tool Performance: Innovations in Reliability and Design

S. Aitmaganbetov, SLB, Dhahran, Eastern province, Kingdom of Saudi Arabia; A. Al Alsayednassir and O. Alvarez, Aramco, Dhahran, Eastern province, Kingdom of Saudi Arabia; V. Radhakrishnan, SLB, Dhahran, Eastern province, Kingdom of Saudi Arabia

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227664-MS](https://doi.org/10.2118/227664-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

This paper delivers a comprehensive evaluation of collaborative engineering efforts aimed at boosting the reliability and efficiency of impact tools, with a specific emphasis on the 9.5-inch drilling jar. The redesigned jar, which features an increased outer diameter, is purpose-built to withstand the extreme downhole conditions typically encountered in top-hole drilling, effectively mitigating issues caused by vibrations and cyclic loading that frequently result in jar malfunctions.

The advanced 9.5-inch jar is a hydraulically actuated, double-acting tool tailored for top-hole applications. It has been carefully engineered to enhance component durability and maintain performance in demanding environments. The design improvements specifically target the reduction of harmful effects from vibration sources such as fluid losses, pack-offs, and elevated bending loads—factors that are commonly linked to drill string failures. This robust tool is intended to elevate service reliability and reduce the likelihood of mechanical breakdowns. Key enhancements include a larger cross-sectional profile, improved resistance to bending, and increased torsional capacity compared to conventional 9.5-inch jars.

Field data from over 113 well deployments, totaling more than 17,120 operational hours, demonstrated zero service quality incidents, highlighting the jar's exceptional reliability and operational performance. The joint engineering initiative among various stakeholders has resulted in a tool that not only meets but exceeds modern performance expectations, significantly improving the dependability and efficiency of drilling operations.

This paper introduces new perspectives on the engineering framework developed to optimize impact tool performance. The jar's innovative design and enhanced operational metrics underscore the value of strategic collaboration in creating high-performance drilling technologies. The results affirm that targeted design enhancements, supported by cross-functional engineering, can lead to substantial gains in tool reliability and operational efficiency.

Introduction

Drilling jars play a vital role in oil and gas operations, particularly when addressing stuck pipe situations. These tools function by delivering high-impact and impulse forces that help dislodge immobilized drill

strings. Their ability to restore movement in the drill string not only enables the continuation of drilling activities but also acts as a critical safeguard for preserving the structural integrity of the bottom hole assembly. The performance of drilling jars is essential to maintaining operational continuity, minimizing downtime, and ensuring the overall cost-efficiency and success of drilling campaigns.

However, one of the most pressing challenges associated with drilling jars is the occurrence of twist-offs, especially in large-diameter boreholes such as 22-inch and 16-inch sections. These failures can lead to substantial non-productive time (NPT), often requiring complex fishing operations or even sidetracking. In extreme cases, they may result in the abandonment of the well. Due to their intricate internal mechanisms and moving components, jars are often considered the most vulnerable segment of the drill string. Their complex geometry introduces stress concentration points, making them particularly susceptible to mechanical failure under high-stress conditions.

The issue of jar twist-offs becomes even more critical in drilling environments that involve transitions across multiple geological formations. Each formation change tends to amplify axial and lateral vibrations, which can induce Bottom Hole Assembly (BHA) whirl. These intensified vibrations place additional mechanical stress on the drill string, particularly on the jars, increasing the likelihood of twist-offs. Compounding the problem are other operational challenges such as severe mud losses and hole enlargement (wash-outs), which further complicate the drilling dynamics and elevate the risk of tool failure.

During a recent drilling campaign, multiple instances of jar twist-offs were documented in the 22-inch and 16-inch hole sections. These incidents accounted for approximately 25–30% of all recorded mechanical failures, highlighting the severity and frequency of the issue. This high rate of occurrence underscores the urgent need for more durable and reliable jar designs capable of withstanding the harsh conditions encountered in large-diameter drilling. Figure 1 provides a visual breakdown of these incidents, illustrating the distribution and impact of jar twist-offs in complex drilling scenarios.

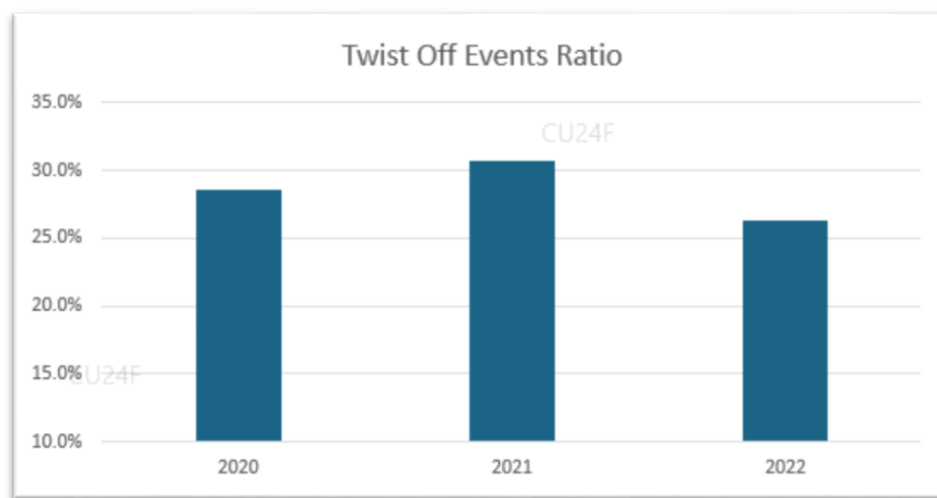


Figure 1—Top hole sections Twist Off Events Ratio

A comprehensive and meticulous evaluation, combined with a detailed investigation of documented twist-off incidents in large-diameter boreholes, has yielded valuable insights into the root causes and recurring patterns of these failures. The in-depth analysis of these events has identified that twist-offs most frequently occur in specific segments of the drill string where there is a distinct transition in cross-sectional geometry—from larger outer diameters to smaller ones. These transition zones act as focal points for stress accumulation, making them particularly prone to mechanical failure under operational loads.

The most susceptible regions have been consistently observed on the Kelly Mandrel and the lower subs of the drilling jar—both of which are critical components within the bottom hole assembly. These areas, due

to their structural configuration and functional role, are exposed to elevated stress concentrations, especially during high-vibration and high-torque drilling conditions. Figure 2 offers a visual depiction of these high-risk zones within the drill string, clearly illustrating the locations where abrupt OD reductions occur and where twist-offs are most likely to initiate.

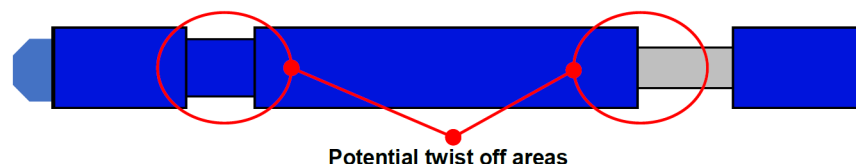


Figure 2—Drilling Jar illustration

This graphical representation reinforces the importance of addressing these vulnerable transition points to enhance the structural integrity and operational reliability of the drilling system. By pinpointing the exact locations and mechanical conditions that contribute to twist-offs, engineering teams are better equipped to implement targeted design modifications and reinforcement strategies. These may include optimizing the geometry of the transition zones, using higher-strength materials, or incorporating advanced fatigue-resistant features.

The findings from this extensive review highlight the critical need for continuous inspection, monitoring, and performance evaluation of drill string components, particularly in large borehole applications where mechanical loads are significantly amplified. Enhancing the durability of the Kelly Mandrel and jar subs through innovative design and material science is essential for reducing the frequency of twist-offs. Moreover, sustained investment in research and development is vital to advancing drilling technologies and methodologies. Such efforts will ensure that the industry remains capable of overcoming the complex challenges posed by cross-sectional transitions and can maintain high levels of efficiency and safety in demanding drilling environments.



Figure 3—Twist off on lower sub



Figure 4—Twist off on Kelly mandrel

A comprehensive series of simulations focused on jar integrity was systematically conducted using actual field Bottom Hole Assembly (BHA) configurations in conjunction with a broad spectrum of drilling parameters. The primary objective of these simulations was to identify specific structural vulnerabilities within the jar components. These virtual tests were carefully designed to replicate realistic

downhole conditions and to encompass a wide array of operational scenarios, thereby enabling a thorough understanding of the mechanical factors that contribute to jar failures during drilling operations.

To achieve this, the research team employed sophisticated computational modeling techniques, including finite element analysis (FEA), to simulate the complex mechanical behavior of the jars under dynamic loading conditions. These simulations allowed for a detailed visualization of stress distribution, deformation patterns, and potential failure points within the jar structure. The BHA configurations used in the modeling process were based on actual field data, ensuring that the simulation outcomes were both accurate and applicable to real-world drilling environments.

Throughout the simulation process, key drilling parameters such as rotational speed, axial force, and fluid circulation rates were methodically varied to assess their influence on the structural integrity of the jars. This approach enabled the identification of critical stress zones and provided insights into how different operational conditions exacerbate mechanical stress within the tool. The results revealed that certain transition areas—particularly those involving changes in cross-sectional geometry—are especially prone to stress concentration and are therefore more susceptible to fatigue and failure.

Figure 5 presents a detailed visual summary of the simulation findings, highlighting the regions within the jar where stress intensities were most pronounced. The illustration emphasizes the importance of these transition zones, where abrupt changes in outer diameter create localized stress peaks. These findings underscore the necessity for targeted design enhancements aimed at reinforcing these vulnerable sections to improve the overall durability and performance of the jar.

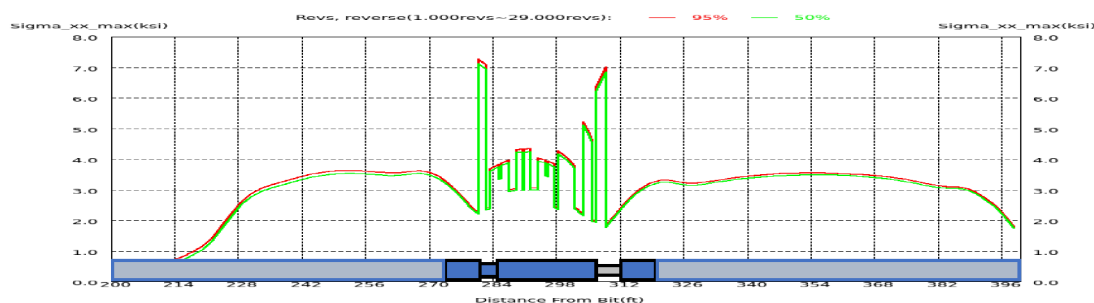


Figure 5—Bending stress analysis across the Jar

The insights gained from this simulation study provide a strong foundation for future design improvements and material selection strategies, ultimately contributing to the development of more robust and reliable drilling jars capable of withstanding the demanding conditions encountered in modern drilling operations.

Objectives/Scope

The primary objective of this in-depth analysis is to substantially improve the operational reliability and performance efficiency of the 9.5-inch drilling jar, particularly in applications involving large-diameter boreholes. This initiative is driven by the persistent challenges posed by extreme downhole environments, which frequently compromise the structural integrity and functionality of drilling tools. By focusing on strategic enhancements to the jar's design, this study aims to ensure that the tool can consistently perform under harsh conditions, thereby minimizing the risk of failure and maintaining operational continuity.

A key emphasis of this research lies in the deliberate refinement of the jar's structural design to better resist the damaging effects of downhole vibrations and cyclic mechanical stresses. These stressors are well-known contributors to fatigue-induced failures and structural degradation within the drill string. Through the application of advanced engineering methodologies and the integration of high-performance materials,

the study seeks to reinforce the jar's resilience, enabling it to endure prolonged exposure to demanding operational loads without compromising its integrity.

In addition, this study underscores the critical value of collaborative innovation across the oil and gas sector. By fostering close cooperation among operators, engineering teams, and technology developers, the research leverages a broad base of expertise and resources to accelerate the development of more robust drilling solutions. These collaborative efforts are expected to yield significant advancements in tool design and performance, ultimately contributing to more efficient and cost-effective drilling operations.

The scope of this work extends beyond technical optimization to embrace a holistic approach to drilling tool development. It emphasizes the integration of real-world field data, simulation outputs, and empirical performance metrics to guide design decisions. By adopting a multidisciplinary framework, the study addresses the complex interdependencies between material behavior, geometric design, and operational parameters that collectively influence tool performance.

In conclusion, the overarching aim of this research is to deliver a transformative improvement in drilling operations by enhancing the durability and efficiency of the 9.5-inch jar. The anticipated outcomes include reduced non-productive time, lower operational risk, and improved resource utilization—key factors in achieving sustainable and high-performance drilling outcomes.

Methods, Procedures, and Process

To effectively reduce the frequency of jar twist-offs, a comprehensive strategy that integrates multiple advanced engineering techniques is essential. One of the most impactful approaches involves upgrading the quality and grade of the metal used in jar construction. By selecting high-performance alloys with enhanced mechanical characteristics, the structural strength and fatigue resistance of the jar can be significantly improved. This material enhancement enables the jar to better endure the extreme mechanical loads and environmental stresses encountered during drilling, thereby lowering the risk of structural failure.

Another critical method centers on optimizing the Bottom Hole Assembly (BHA) through precise and strategic placement of the jar. Proper positioning within the BHA allows for more balanced force distribution and minimizes the development of localized stress concentrations. This optimization requires a detailed evaluation of drilling parameters and operational dynamics to determine the most effective location for the jar. When placed optimally, the jar operates in a more stable mechanical environment, which contributes to improved performance and a reduced likelihood of twist-offs.

Additionally, adjusting surface parameters to reduce stress concentrations is another technique that can be employed to mitigate the risk of jar failure. However, it is important to note that while minimizing surface parameters can help reduce mechanical stress, it may also lead to a decrease in overall drilling efficiency, presenting a trade-off that must be carefully managed.

Despite the effectiveness of these mitigation strategies, the fundamental vulnerability of the jar remains unless its core design is reengineered. To address this, a comprehensive redesign of the jar is recommended, focusing on enhancing its structural robustness under similar operational loads. This involves modifying the jar's geometry to improve its mechanical performance and resistance to fatigue.

As shown in [Figure 6](#), a key improvement in the redesigned jar is the increase in the diameter of the critical cross-sectional area—from 6.00 inches to 7.01 inches. This change results in a significant increase in cross-sectional area, expanding from 18.55 square inches to 28.09 square inches. The added metal volume in these critical regions strengthens the Kelly mandrel, making it more capable of withstanding high stress and reducing the risk of twist-offs.

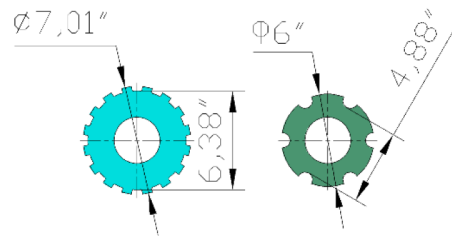


Figure 6—Splined Mandrel cross section

Figure 6 visually illustrates these design modifications, clearly highlighting the enhanced cross-sectional dimensions and increased metal concentration. These improvements are essential for boosting the jar's structural integrity and ensuring reliable performance in demanding drilling environments.

A detailed Finite Element Analysis (FEA) was systematically performed on the current design of the 9.5-inch drilling jar and subsequently compared with a newly optimized version. This advanced simulation technique was utilized to gain a comprehensive understanding of the structural behavior and mechanical performance of both designs under a variety of operational conditions. The FEA process incorporated a wide range of critical variables, including material characteristics, applied loads, and geometric configurations, to ensure a robust and realistic evaluation.

The simulated loading conditions were carefully selected to reflect the actual forces encountered by the jar during drilling operations. These included axial compression and tension, torsional shear, and bending moments, all of which are commonly experienced in downhole environments. By replicating these real-world mechanical stresses, the analysis provided a realistic representation of the jar's performance, allowing for a meaningful comparison between the existing and improved designs.

Geometric features played a pivotal role in the FEA simulations, as the shape, dimensions, and transitions within the jar directly influence how stress is distributed throughout the structure. The analysis examined detailed aspects of the jar's geometry, such as cross-sectional profiles, tapering zones, and surface contours. By comparing these features across both designs, engineers were able to pinpoint areas where stress concentrations were most likely to develop. These insights were essential in guiding the redesign process to improve structural resilience.

The optimization phase focused on redistributing mechanical loads more uniformly across the jar's body to reduce localized stress peaks. This was achieved through targeted design modifications, including increasing the cross-sectional area and refining geometric transitions to create smoother load paths. These enhancements aimed to reduce the likelihood of fatigue failure and improve the jar's ability to withstand prolonged exposure to harsh drilling conditions.

The comparative results from the FEA simulations revealed that the new design outperformed the original in several key areas, including stress distribution, load-bearing capacity, and overall structural integrity. The improved design demonstrated a more balanced response to applied forces, significantly lowering the risk of mechanical failure. These findings validate the effectiveness of the design improvements and highlight the value of using FEA as a powerful tool in the development of high-performance drilling equipment.

As depicted in Figure 7, a comparative stress analysis between the original and the enhanced jar designs under identical loading conditions reveals a substantial improvement in structural performance. The refined design demonstrated a maximum stress level of 50 Kilo Pounds per Square Inch (ksi), representing a significant reduction from the 95 ksi recorded in the original configuration. This considerable decrease in stress levels clearly reflects the effectiveness of the design modifications in improving the jar's mechanical resilience and overall structural integrity.

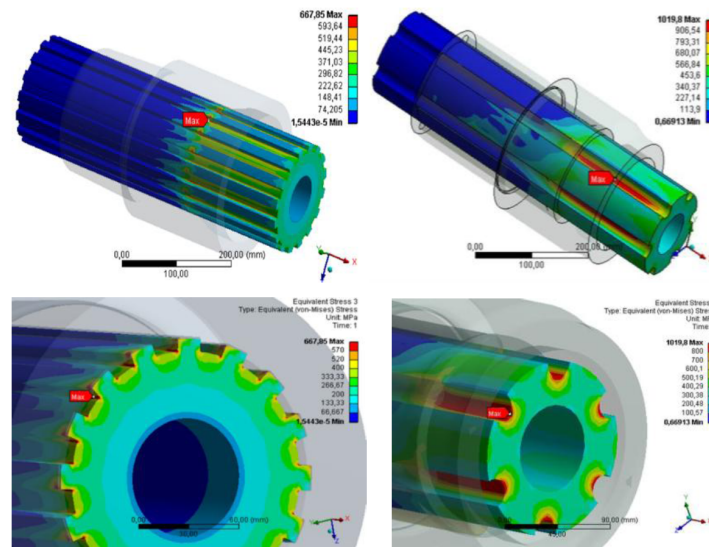


Figure 7—FEA analysis comparison

One of the most notable advancements in the improved design is its enhanced resistance to bending stress, which has effectively been doubled. This improvement is particularly critical in drilling environments where bending forces are frequently encountered. The increased bending resistance ensures that the jar maintains its structural stability and operational functionality even under severe bending loads. This enhancement is a direct result of strategic changes in the jar's geometry and material distribution, which collectively contribute to its improved durability.

Under equivalent torsional loading conditions, the improved design exhibited a stress level of 97 ksi, a significant reduction from the 148 ksi observed in the original design. This reduction highlights the superior torsional strength of the new configuration, which is essential in withstanding the rotational forces commonly experienced during drilling. The improved torsional performance is attributed to the increased cross-sectional area and the strategic concentration of material in high-stress regions, which collectively reinforce the jar's structural framework.

The increase in cross-sectional area and the redistribution of metal in the improved design have also led to notable enhancements in both the tensile and torsional yield limits. The tensile yield limit defines the maximum axial stress the jar can endure without undergoing permanent deformation, while the torsional yield limit refers to the maximum shear stress it can withstand before structural failure occurs. The elevated yield thresholds in the improved design ensure that the jar can operate reliably in high-stress drilling environments, where mechanical demands are consistently intense.

Figure 7 provides a visual comparison of the stress profiles for both designs, clearly illustrating the areas where stress has been significantly reduced in the improved version. The graphical representation emphasizes the benefits of the design enhancements, particularly in critical cross-sectional zones. By reinforcing these areas and optimizing the distribution of material, the improved design achieves a more balanced and robust structural configuration, ultimately reducing the risk of twist-offs and enhancing operational reliability.

Field Testing

A total of five trial runs were successfully carried out using the newly engineered jar design across different wellbore sections.

- **1st Run:** The newly developed 9-1/2" jar was deployed in the 22" hole section, covering a depth of over 2,000 feet. Despite encountering severe drilling conditions caused by total fluid losses, the jar

exhibited no signs of cracking or fatigue-related damage. Its successful performance under such harsh circumstances highlights the robustness of the design and its ability to maintain structural integrity in challenging formations.

- **2nd Run:** The enhanced 9-1/2" jar was utilized in the 16" section, drilling more than 2,800 feet. Throughout this extended interval, the jar delivered consistent and reliable performance, reinforcing its durability and operational dependability over prolonged use in demanding environments.
- **3rd Run:** In this instance, the 9-1/2" jar was again employed in a 22" section, successfully drilling over 1,600 feet without any service quality issues. The jar maintained full functionality and structural soundness throughout the operation, further validating its mechanical strength and endurance.
- **4th Run:** The jar was used to drill a 16" section exceeding 3,300 feet. During the operation, the Bottom Hole Assembly (BHA) became stuck while being pulled out of the hole, requiring 11.5 hours of intervention, including acid pumping through the drill string. Despite the stuck pipe event, the jar continued to function effectively, demonstrating its resilience and ability to perform under prolonged stress and adverse conditions.
- **5th Run:** The jar was deployed in a 22" section, drilling over 2,140 feet. High rotational speeds and elevated Weight on Bit (WOB) were applied, achieving a penetration rate of 65 ft/hr—an improvement of 95.2% compared to previous runs. The jar performed exceptionally well under these optimized parameters, showcasing its capability to enhance drilling efficiency and withstand aggressive drilling conditions.

As illustrated in Figure 8, the performance efficiency of impact tools has shown a steady year-over-year improvement following the adoption of one of the key enhancement strategies. Notably, this upward trend in efficiency has persisted despite a consistent 9% annual increase in total operational time. This indicates that the tools are not only being used more extensively but are also becoming more effective over time. A major contributor to this sustained improvement is the redesign of the drilling jar, which has played a pivotal role in boosting tool reliability and operational output.

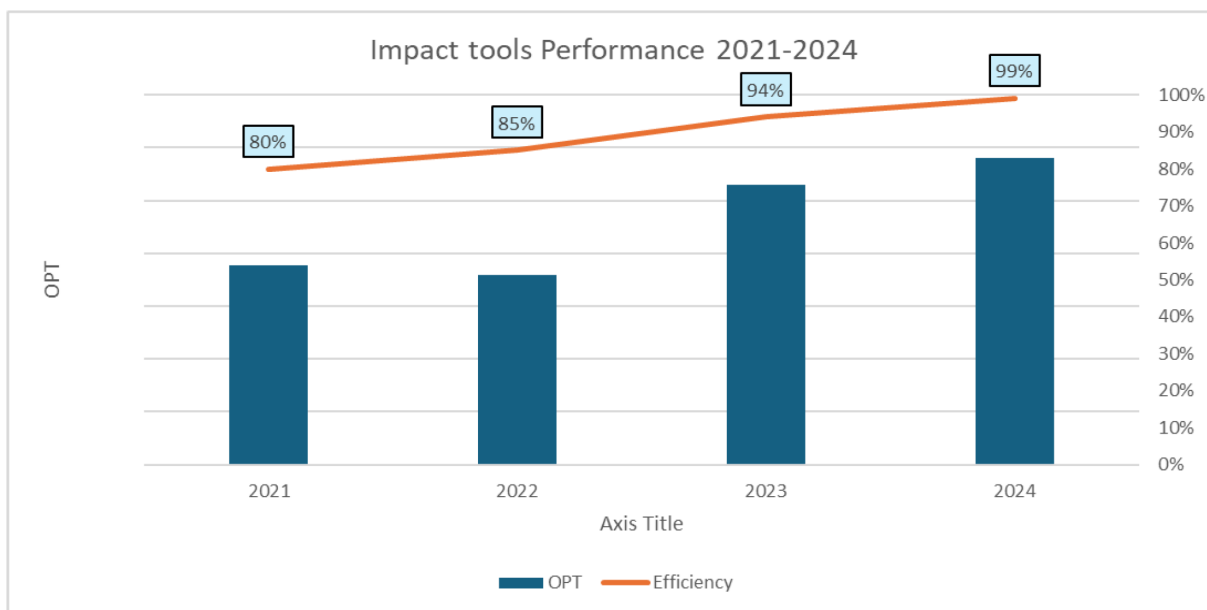


Figure 8—Impact Tools Efficiency Improvement

Conclusion

The introduction of the newly engineered 9.5-inch hydraulic double-acting drilling jar marks a major leap forward in the evolution of impact tool technology. This innovative design incorporates several key enhancements that significantly improve the jar's mechanical performance and operational reliability:

- **Expanded Cross-Sectional Area:** The jar's cross-sectional area has been increased by 39%, resulting in a more robust structure capable of withstanding higher mechanical loads and reducing the likelihood of structural failure.
- **Greater Bending Strength:** The tool's resistance to bending forces has been improved by 47%, making it far more resilient in scenarios where lateral stresses are prevalent, such as in deviated or complex wellbores.
- **Improved Torsional Capacity:** With a 34% increase in torsional strength, the jar is now better equipped to handle rotational stresses, significantly lowering the risk of twist-offs during high-torque operations.

These design advancements collectively enhance the jar's ability to perform reliably in severe downhole environments, where traditional tools often face limitations. The successful deployment of this new jar design in more than 113 wells—without a single reported service quality issue—demonstrates its effectiveness and underscores the importance of strategic collaboration in driving technological innovation within the drilling industry.

References

1. Oliveira, C., Martinez, A., S. Zahrani, S., Abouelnaaj, K. 2017. A Study of Drilling Jar Utilization and Effectiveness in the Kingdom of Saudi Arabia. Paper SPE-185364 presented at the SPE Oil and Gas India Conference and Exhibition, India, 4–6 April.
2. Boufama, C., Yang, B., Alfaraedhi, S., 2022. Service Quality Improvement Plan Engineered Using FEA Fatigue Analysis Minimizes Risk of Jar Twist-off in One of the Most Challenging Applications. Paper SPE-210723 presented at SPE Asia Pacific Oil & Gas Conference and Exhibition, Adelaide, Australia, 17–19 October.
3. Herrington, D., Barton, S., Bouaziz, S., Clausen, J. 2012. Step Change Jar Technology Redefines Industry Standard in North Sea Region. Paper SPE 160428 presented at the SPE Asia Pacific Oil and Gas Conference, Perth, Australia, 22–24 October.
4. Lerma, M.K. 1985. A Study of How Heavy-Weight Drill pipe Affects Jarring Operations. Paper SPE 14326 presented at the 60th Annual Technical Conference and Exhibition, Las Vegas, Nevada, 22–25 September.
5. Mercado, J. L. 2013. Planning for Successful Jarring Operations – Effective Use of Drilling Impact System Helps Release Stuck Pipe. Paper SPE/IADC 163413 presented at the IADC/SPE Drilling Conference and Exhibition, Amsterdam, The Netherlands, 5–7 March.
6. Mario A. Rivas, Andres A. Ramirez, Bader S. Al-Zahrani, and Khaled K. Abouelnaaj. 2021. Mitigating Twist-Offs While Drilling with the Help of BHA Dynamics Software. Paper SPE-202174 presented at the SPE/IADC Middle East Drilling Technology Conference and Exhibition, Abu Dhabi, UAE, 25 – 27 May 2021.